IOC Enrichment Agent (Agent 7)

Deep Technical Specification & Operational Manual

Executive Overview & Purpose

The IOC Enrichment Agent takes raw indicators of compromise extracted by triage and enriches them against external threat intelligence platforms (VirusTotal, Shodan) and local deterministic reputation heuristics, producing a consolidated, risk-weighted IOC registry.

1. Architectural Overview & Workflow

- Position in LangGraph Engine: Second node in Pipeline B, executing directly after IncidentTriageAgent.
- Input Consumption: Consumes state.extracted_iocs (IPs, domains, hashes, URLs).
- State Propagation: Writes state.ioc_enrichment_report and updates findings with per-IOC reputation context.
- Graceful Degradation: Never fails the pipeline when external APIs are unreachable; deterministic heuristics fill the gap.

2. Multi-Source Enrichment Layer

- VirusTotal API v3: Reputation votes (malicious/suspicious/harmless), country attribution, and tags per IOC.
- Shodan REST API: Open-port inventory, ISP, hostnames, and known vulnerabilities for exposed IPs.
- Local Deterministic Heuristics: Private-range classification, reserved multicast/loopback detection, hash format validation, and static reputation scoring.

3. Risk-Weighted IOC Registry

- Threat Score: Each IOC receives a normalized 0-100 risk score combining external votes and local heuristics.
- Confidence Labeling: IOCs are labeled MALICIOUS, SUSPICIOUS, or BENIGN with evidence from each contributing source.
- Deduplication: Consolidates repeated indicators while preserving first-seen/last-seen provenance.

4. Output Schema & Downstream Handoff

- IOCEnrichmentReport: Contains enriched_iocs list, threat_summary, and overall threat_score.
- Downstream Consumption: LogCorrelationAgent and ForensicsAgent consume enriched IOC context for behavioral correlation.